

CELL BIOLOGY

AI-assisted protein design to rapidly convert antibody sequences to intrabodies targeting diverse peptides and histone modifications

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Intrabodies are engineered antibodies that function inside living cells, enabling therapeutic, diagnostic, and imaging applications. While powerful, their development has been hindered by challenges associated with their folding, solubility, and stability in the reduced intracellular environment. Here, we present an artificial intelligence–driven pipeline integrating AlphaFold2, ProteinMPNN, and live-cell screening to optimize antibody framework regions while preserving epitope-binding complementarity-determining regions. Using this approach, we successfully converted 19 of 26 antibody sequences into functional single-chain variable fragment intrabodies, including a panel targeting diverse histone modifications for real-time imaging of chromatin dynamics and gene regulation. Notably, 18 of these 19 sequences had failed to convert using the standard approach, demonstrating the unique effectiveness of our method. As antibody sequence databases expand, our method will accelerate intrabody design, making their development easier, more cost effective, and broadly accessible for biological research.

INTRODUCTION

Intrabodies are a special class of antibodies that are engineered to fold and function within the intracellular environment (1, 2). This property gives intrabodies great therapeutic and diagnostic potential because they can bind target biomolecules deep within living cells and tissue. Furthermore, when intrabodies are tagged with fluorescent proteins, they become powerful imaging reagents that can bind and light up challenging targets that cannot be imaged with standard genetic fusion tags such as green fluorescent protein (GFP). For example, fluorescent intrabodies have been used to image conformational (3–6) and linear epitopes (7, 8), macromolecular complexes (9, 10), and chemical modifications (11–13).

Despite the broad applicability of intrabodies, engineering them has proven to be a substantial challenge. The antibody scaffolds from which intrabodies are derived have evolved to function in extracellular environments such as the bloodstream, where conditions support the formation of disulfide bonds that are critical for maintaining structure. In contrast, the reduced intracellular environment disrupts these bonds, often leading to issues such as misfolding,

aggregation, and degradation (14–16). In addition, the intracellular environment features numerous differences from the extracellular environment in terms of off-target binding and aggregation. For these reasons, the straightforward placement of antibody domains into an intrabody format—whether it be a single-chain variable fragment (scFv) (17) or a nanobody (2, 18–20) (Table 1)—often results in a nonfunctional protein. Thus, despite the wide availability of antibodies commercially and in protein databases, there are relatively few functional intrabodies (21, 22).

One specific area of research that could benefit from additional intrabodies is the study of histone modifications (11). These reversible chemical marks—which include acetylation, methylation, and phosphorylation, among others—decorate the unstructured tails of nucleosomes across the genome. The modifications are believed to work both individually and in combination to help shape chromatin architecture and regulate gene expression (23, 24). Hence, they are strongly correlated with human health and disease, making them promising therapeutic targets (25, 26). However, because the marks are reversible, they are difficult to image in living cells using standard genetic tags (27, 28). Instead, specialized imaging probes such as intrabodies are needed that can dynamically bind and light up the modifications as they come and go. Although there has been some progress developing such probes (27, 28), the pace has been slow due to the inherent engineering challenges. For this reason, the dynamics of histone modifications have remained elusive, as has their precise roles in gene regulatory networks.

Recent advances in artificial intelligence (AI), including tools such as AlphaFold2 (29), RFdiffusion (30), and protein large language models (31), offer transformable opportunities to overcome these challenges. These technologies not only enable the de novo creation of proteins with functions not attainable by natural sequences (32–34) but also provide a powerful platform to improve existing proteins such as antibodies. While numerous efforts have focused on optimizing the complementarity-determining regions of antibodies

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Table 1. Terminology and abbreviations.	
Term	Definition
scFv	Single-chain variable fragment
Intrabody	Antibody-based probe (like scFv) that functions in live cells
Mintbody	Intrabody that targets posttranslational protein modifications
Nanobody	Single-domain antibody-based probe derived from camelids
Fab	Antigen-binding fragment
Frankenbody	An engineered scFv created by grafting CDR loops onto a stable scFv framework so it folds and functions in living cells
CDR	Complementarity-determining region
FLAG tag	Short peptide epitope tag (sequence DYKDDDDK) used for detection/purification of tagged proteins
HA tag	Short peptide epitope tag (sequence YPYDVPDYA) used for detection/purification of tagged proteins
H2B	Histone 2B, one of the core histone proteins that package DNA into nucleosomes
PTM	Posttranslational modification: covalent modification to a protein after translation (e.g., phosphorylation, acetylation, methylation)
Histone PTM notation	Notation for histone posttranslational modifications; example: H3K27ac = Histone H3, lysine-27 acetylation
GFP	Green fluorescent protein
FRAP	Fluorescence recovery after photobleaching
FCS	Fluorescence correlation spectroscopy
TSA	A histone deacetylase (HDAC) inhibitor used experimentally to increase histone acetylation levels
A-485	A selective inhibitor of p300/CBP histone acetyltransferases used to reduce histone acetylation experimentally
p300	A histone acetyltransferase (HAT) that acetylates lysine residues (including H3K27), targeted by inhibitors like A-485.
UNC0642	Small-molecule inhibitor of G9a/GLP histone methyltransferases
DS3201	Small-molecule inhibitor of EZH1/2 histone methyltransferases
ZM447439	Aurora B kinase inhibitor; blocks phosphorylation of H3S10/S28
CHR6494	Haspin inhibitor; blocks phosphorylation of H3T3
PCNA	Proliferating cell nuclear antigen; a protein that marks replication foci and is used as a marker for DNA synthesis during S phase
pLDDT	Predicted local distance difference test (confidence metric from AlphaFold)
pTM	Predicted template modeling score (confidence in overall topology)
PAE	Predicted aligned error (per-residue error estimate in AlphaFold models)
SWI	Solubility weighted index (predicted solubility metric)
pK _a	Logarithm of the acid dissociation constant of ionizable groups in proteins

to improve target binding affinity (35–37) or the optimization of therapeutic antibody sequences to minimize self-association and off-target binding under standard formulation conditions (38, 39), it remains unclear whether these technologies can similarly enhance the framework regions of antibodies to improve their functionality when expressed as intrabodies in living cells.

To address this issue, we here develop an AI-driven approach to accelerate the development of functional intrabodies. Using a novel pipeline that leverages antibody domain annotation (40), AlphaFold2 (29, 41), ProteinMPNN (42), and live-cell screening, we demonstrate how existing antibody sequences can be rapidly converted into functional intrabodies. As a proof of principle, we first develop two peptide-binding intrabodies, one targeting the synthetic FLAG tag (43) and the other targeting a linear epitope of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) nucleocapsid protein (NP) (44). We then turn our attention to endogenous histone modifications. To highlight the speed and practicality of our approach, we create a panel of intrabodies that bind diverse residue-specific chemical modifications on both histones 3 (H3) and 4 (H4), including H3 threonine-3 phosphorylation (H3T3ph), H3 lysine-4 monomethylation (H3K4me1), H3 lysine-4 dimethylation (H3K4me2), H3 lysine-4 trimethylation (H3K4me3), H3 lysine-9 (un)methylation [H3K9(un)],

H3 serine-10 phosphorylation (H3S10ph), H3 lysine-14 acetylation (H3K14ac), H3K27me1, H3K27me2, H3K27ac, H3K36me3, H4K5ac, H4K8ac, H4K12ac, and H4K16ac, tripling the number of intrabodies available for this purpose (27). We demonstrate the functionality of these intrabodies in living cells and provide their sequences so they can be further refined or modified to image or manipulate histone modifications in vivo. Furthermore, we provide open-source code implementing our pipeline and provide useful metrics and tables to help predict which candidate designs will retain functionality inside cells. As the number of published antibody sequences continues to rapidly grow, we believe that our approach will substantially increase the number, utility, and accessibility of intrabodies to scientists in the life sciences.

RESULTS

Developing a pipeline for transforming existing antibody sequences into enhanced scFvs for intracellular applications

We developed a pipeline to explore the potential of AI-based protein design tools to enhance scFv engineering (Fig. 1A). We were driven by failures to produce a functional scFv based on the commercial M2 anti-FLAG antibody, whose peptide sequence had recently been

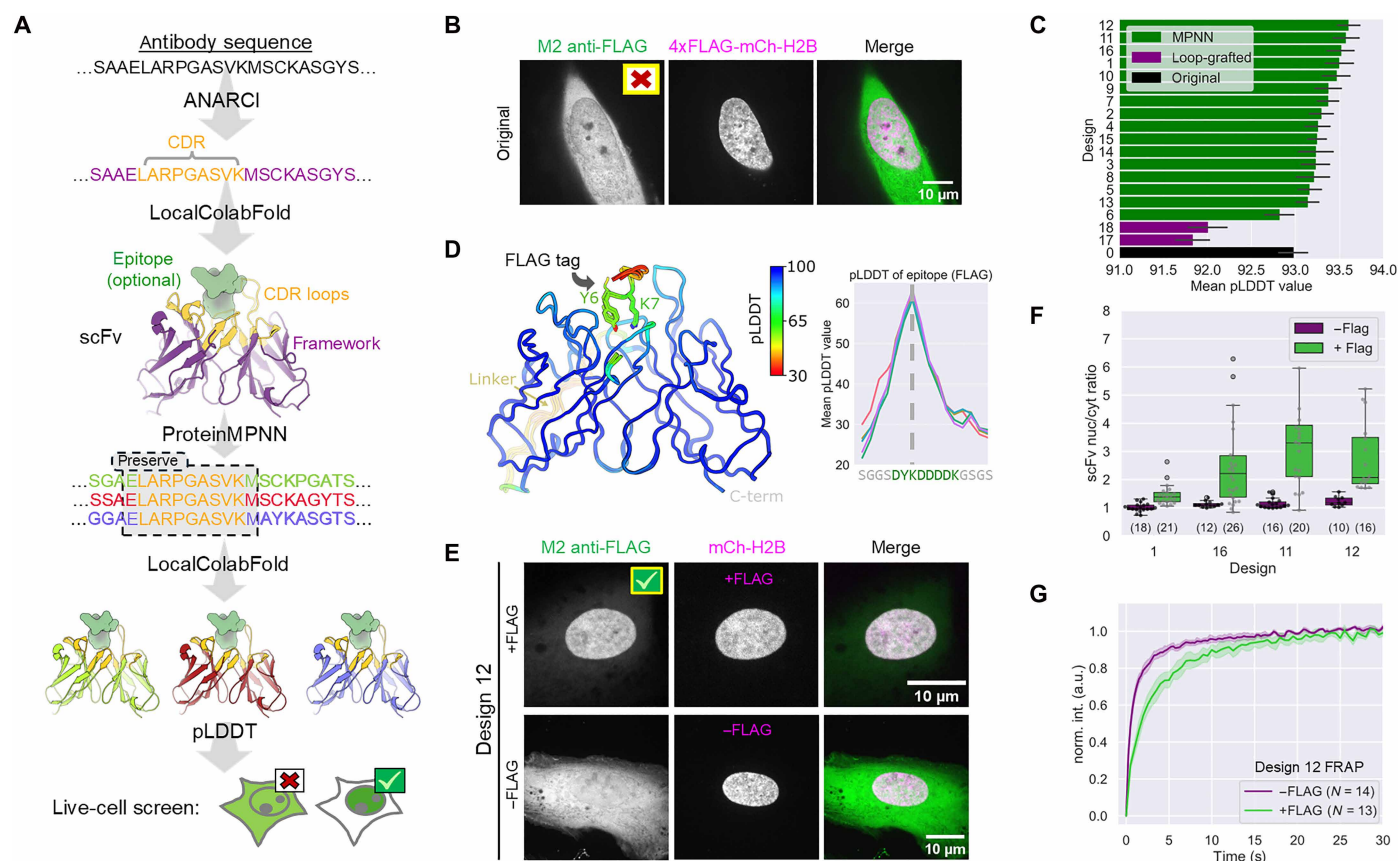


Fig. 1. Pipeline to convert antibody sequences into functional intrabodies. (A) Schematic showing the pipeline, from antibody sequence to live-cell screening. The pipeline relies on three software platforms: (i) ANARCI (antigen receptor numbering and receptor classification), a tool to classify and number antibody amino acids; (ii) LocalColabFold, an offline, local version of ColabFold that accelerates AlphaFold-base protein structure predictions by integrating a search tool MMseqs2; and (iii) ProteinMPNN (protein message passing neural network), a deep-learning method for protein design applicable to protein stability and solubility optimization. (B) The original M2 anti-FLAG scFv (green) does not colocalize with 4xFLAG-mCh-H2B (magenta) in the nucleus. (C) The ranked predicted local distance difference test (pLDDT) scores of all M2 anti-FLAG scFv designs (include scFv and FLAG epitope), including the original (black bar), two loop-grafted variants (purple), and 16 from the pipeline (green). Error bars show the SD from LocalColabFold-predicted models ($N = 5$ each). (D) The top four predicted structures (models 1, 16, 11, and 12) overlaid with the original M2 anti-FLAG structure (model 0), colored by pLDDT. Right: The corresponding overlaid pLDDTs of the tested binder, with the FLAG tag (DYKDDDDK) indicated in green text. C-term, C-terminal. (E) Sample cell from the live-cell screen of design 12. This scFv (green) colocalizes with the FLAG-tagged H2B (magenta). When there is no FLAG-tagged protein coexpressed, the scFv shows uniform localization (bottom). (F) Live-cell measurement of the N/C ratio of the indicated scFv designs [number of cells (N) indicated; box is 25 to 75% interquartile range; whiskers mark points within $1.5\times$ the interquartile range]. nuc/cyt, nuclear/cytoplasmic. (G) Fluorescence recovery after photobleaching (FRAP) experiments in the nucleus of cells expressing design 12 scFv in the presence and absence of FLAG-H2B. The number of cells (N) is indicated; the shaded region shows the 95% confidence interval (CI). norm. int., normalized intensity. a.u., arbitrary units.

published (45). Initially, we reformatted the M2 sequence into a standard scFv with a flexible glycine-serine (GS)-rich linker. However, when fused to GFP and coexpressed with FLAG-tagged histone 2B (H2B), the scFv failed to colocalize with its target, instead showing uniform cellular distribution—a clear indication of it remaining unbound and freely diffusing (Fig. 1B). We attempted to fix this problem by loop grafting the M2 complementarity-determining regions (CDRs) onto a stable scFv framework region that we had previously used to generate anti-hemagglutinin (HA) (7) and anti-FLAG (46) “frankinbodies.” We also tested a new scFv framework region derived from an anti-polymerase II scFv (47), selected because of its high sequence similarity to the M2 frameworks. Unfortunately, none of these variants achieved specific colocalization with FLAG-tagged H2B (fig. S1A).

Given the widespread use of the M2 antibody, we were convinced that our scFv failed in living cells due to misfolding or insolubility, rather than a failure in antigen binding and specificity. We hypothesized that if we could design more soluble scFv framework regions based on the original sequence, then this would enhance their compatibility with the original CDRs, thereby increasing the likelihood of successful loop grafting. To test this hypothesis, we created a computational pipeline that integrates several AI-based tools. The pipeline begins by annotating the antibody sequence to extract the CDRs and framework regions using ANARCI (antigen receptor numbering and receptor classification) (40). Second, we predict the three-dimensional (3D) structure of the scFv:target complex using LocalColabFold (41), which combines AlphaFold2 (29) with the fast multiple sequence generation method MMseqs2 (48, 49). Third, we

input the predicted structure into ProteinMPNN (42), holding regions near the annotated CDRs fixed. For the remaining residues, ProteinMPNN attempts to predict favorable new amino acid selections. Crucially, the resulting ProteinMPNN framework sequence differs from the original but tends to encode a protein that is more well behaved because the underlying neural network has been trained on well-behaved, soluble proteins with solved crystal structures (42, 50). We hypothesized that these new framework sequences would be preferable with respect to combating aggregation of the designed proteins while maintaining the critical 3D structure needed for compatibility with the original CDRs.

Running the M2 anti-FLAG scFv through the pipeline resulted in 15 designs. Compared to the original M2 anti-FLAG scFv and the two unsuccessful loop-grafted versions, the new designs exhibited considerably improved pLDDT (predicted local distance difference test) scores (Fig. 1C). AlphaFold2-predicted 3D structures of these designs closely matched the original model (Fig. 1D, left). While the per-residue confidence (pLDDT) was substantially lower for the FLAG epitope than the scFv (Fig. 1D) (excluding the linker), we noted that the confidence in the authentic epitope was nonetheless higher than the confidence for flanking residues, hinting at latent detection of binding specificity by AlphaFold2. Notably, the pLDDT scores for the FLAG epitope were higher than those for adjacent residues (Fig. 1D, right), even when the FLAG epitope was challenged with a nearby HA epitope (fig. S1B), indicating binding specificity. Furthermore, the predicted orientation of the FLAG epitope matched that of a recently determined structure based on the full-length parental M2 antibody (8RMO) (51), despite being predicted before the deposition of the structure in the Protein Data Bank (fig. S1C).

We next screened the top four designs in living cells, selected on the basis of pLDDT scores (designs 12, 11, 16, and 1 in Fig. 1C). All candidates successfully colocalized with FLAG-H2B in the nucleus, as quantified by the ratio of nuclear to cytoplasmic (N/C) fluorescence (Fig. 1, E and F). The specificity of the best scFv (design 12) was further validated in cells coexpressing a FLAG-tagged mitochondrial protein, in which case the scFv now colocalized with mitochondria (fig. S1D). In contrast, in cells without FLAG tags, the scFv displayed homogenous localization, indicating minimal off-target binding [Fig. 1, E (bottom) and F (purple)]. In agreement with these results, FLAG-tagged mCherry (mCh) was efficiently coprecipitated with the GFP-tagged scFv by anti-GFP beads from doubly transfected human embryonic kidney (HEK) 293T cell lysates (fig. S1E). Live-cell fluorescence recovery after photobleaching (FRAP) experiments on design 12 further confirmed specific binding, being slower in the presence of FLAG-H2B (Fig. 1G), with a recovery half-time of ~2 s. The timescale of this recovery is similar to FRAPs we performed on monovalent antibodies (Fab) generated from full-length antibodies with affinities in the 10 nM range (52). Consistent with this comparison, the affinity of the full-length M2 anti-FLAG antibody was recently estimated to be ~2 nM (51). These data therefore suggest the CDRs of our M2 scFvs remain intact despite the large number of changes we made to their framework regions. This underscores the success of our pipeline in generating functionally improved scFvs that maintain specific antigen recognition.

Using the pipeline to create an intrabody against the SARS-CoV-2 NP

To determine whether our technique could be generalized, we next targeted a medically relevant antigen: the NP of SARS-CoV-2. The

NP is a promising diagnostic target due to its early and abundant expression during viral infection (53). We had previously developed and sequenced an antinucleocapsid mouse monoclonal antibody (mBG17), identifying its nucleocapsid epitope (EpNP) as a short 10-amino acid peptide (DDFSKQLQQS) that forms an α -helical structure (44, 54). Similar to our experiences with the M2 anti-FLAG scFv, when we attempted to convert this sequence into an intracellularly functional scFv or frankenbody, we were unsuccessful (fig. S2A). These previous failures made the anti-EpNP scFv an ideal candidate to further test our computational pipeline. Furthermore, unlike the FLAG epitope and M2 anti-FLAG antibody, which are well characterized and supported by crystal structures available in the Protein Data Bank, the mBG17 antibody lacks structural information, offering an unbiased test case for evaluating the predictions of our pipeline.

Mirroring our work with the M2 anti-FLAG antibody sequence, we generated 45 new scFv sequences targeting the NP epitope. All had higher pLDDT scores than the original, with some designs better than others (fig. S2B). To assess how predictive these pLDDT scores were for live-cell functionality, we focused on six designs across the pLDDT score range (Fig. 2B). The predicted 3D structures for these six designs showed consistent placement of the epitope within the antigen-binding pocket (Fig. 2C), with the central portion having especially high pLDDT values (Fig. 2C). This aligned well with our earlier work (44) that identified the key residues within the NP epitope that are critical for binding to the antibody, bolstering our confidence in the predicted structures.

To screen individual designs, we cotransfected each in cells together with a plasmid encoding a 4 $\times$ EpNP-tagged H2B. Three of the six tested anti-EpNP scFv designs successfully colocalized with the EpNP-H2B in the nucleus (Fig. 2, D and E). The successful candidates were not those with the highest pLDDT scores, suggesting that high scores alone might not always predict solubility and functional success. To verify specific binding, we expressed the successful scFv in cells lacking H2B-EpNP. In these cells, we observed a uniform distribution, indicating that the scFv was freely diffusing and unbound, with no detectable off-target binding [Fig. 2, D (lower) and E (purple)]. FRAP further confirmed this specificity, showing slower recovery rates in cells containing H2B-EpNP compared to those without it (Fig. 2F).

In our analysis of the sequence alignment of the screened scFv variants, we identified a feature that correlated with success: Residue 80 was lysine (K) in more effective designs versus serine (S) or asparagine (N) in less effective designs (fig. S2, C and D). This residue is situated in a β hairpin turn outside the CDR loops and could play a role in stabilizing CDR conformations (fig. S2D). Our previous studies that used random mutagenesis on different scFvs also identified residue 80 as crucial for scFv solubility (55). However, the presence of K80 alone is not sufficient for enhanced performance, as the original sequence also contained K80. This suggests that the improved solubility and functionality likely result from modifications to multiple residues. Nonetheless, these observations highlight the effectiveness of our strategy in rapidly generating functional scFv sequences and also demonstrates how a comparison of different designs can be used to quickly generate hypotheses about key residues.

Developing intrabodies targeting posttranslational H3 modifications

Encouraged by these successes, we next set out to develop intrabodies that target endogenous posttranslational histone modifications.

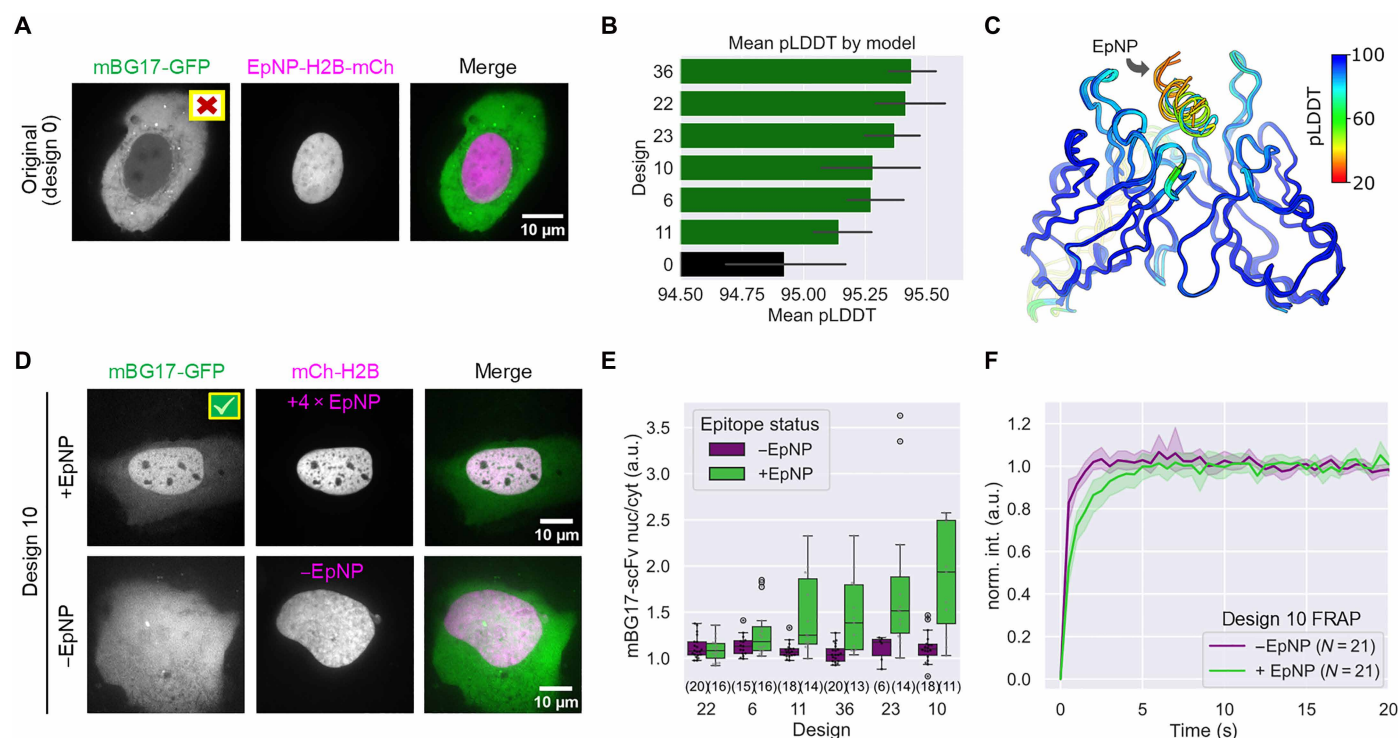


Fig. 2. Engineering a functional intrabody against the NP from SARS-CoV-2. (A) An scFv (mBG17-GFP; green) based on the sequence of a wild-type antibody against the SARS-CoV-2 NP does not colocalize with its target epitope (EpNP-H2B-mCh; magenta) in living cells. (B) The ranked pLDDT scores of a selection of mBG17 pipeline designs (includes both scFv and EpNP; green bars) along with the original (black bar). Error bars show the SD from LocalColabFold-predicted models ($N = 5$ each). (C) The predicted structures (models 11, 6, 10, 23, 22, and 36) overlaid, colored by pLDDT. (D) Design 10 (green) colocalizes with its target epitope (top; +EpNP = EpNP-mCh-H2B; magenta) and displays uniform fluorescence throughout cells in the absence of its target epitope (bottom; -EpNP = mCh-HB; magenta). (E) Live-cell measurement of the N/C ratio of the indicated scFv designs [number of cells (N) indicated; box is 25 to 75% interquartile range; whiskers mark points within $1.5\times$ the interquartile range]. (F) FRAP experiments in the nucleus of cells expressing design 10 mBG17 scFv in the presence (+EpNP = EpNP-mCh-H2B) and absence (-EpNP = mCh-H2B) of EpNP. The number of cells (N) is indicated; the shaded region shows the 95% CI.

We refer to such modification-specific intracellular antibodies as “mintbodies” (13, 27). Mintbodies dynamically bind their target modifications, enabling real-time imaging and tracking in living cells without affecting cell divisions and animal development (27, 56).

Since 2013, five mintbodies against histone and RNA polymerase II modifications have been developed (13, 47, 55, 57, 58). The first two were selected from several hybridoma clones (13, 57) and the other three involved random and/or site-directed mutagenesis with extensive screening (47, 55, 58). In the latter cases, the original scFvs were only a few mutations away from being functional in the intracellular environment, suggesting they may have been fortuitously close to working from the start. On the basis of our experience with anti-EpNP, we suspect that other scFvs may require many mutations before they can properly fold and function inside cells, in which case random mutagenesis becomes impractical due to the immense combinatorial possibilities. We therefore hoped that our computational pipeline could substantially reduce the search space and expedite the production of useful mintbodies.

To test this, we decided to focus on a set of mouse monoclonal antibodies targeting three distinct modifications to H3: K27ac (anti-H3K27ac, clone 9E2H10/CMA309) (59), S10ph (anti-H3S10ph; clone 7G1G7/CMA311) (60), and K9me3 (anti-H3K9me3; clone 2F3/CMA318) (61). These antibodies are excellent candidates for our pipeline because—despite being sequenced and well characterized—each

fails to bind its target modification in living cells when converted into an scFv format (Fig. 3A). Furthermore, the structures of these antibodies have not been experimentally determined, so our pipeline predictions should be free from any bias that could arise from AlphaFold having prior access to the data during its training.

For each antibody sequence, we generated 15 candidate scFvs with our pipeline and chose the top four or five by pLDDT to screen in living cells. To see whether modeling the target epitope was critical, we tested a couple of strategies. For H3K27ac, we input the scFv along with the target peptide (the H3 N-terminal tail with glutamine at position 27 to mimic acetylation), while for H3S10ph and H3K9me3, we just input the scFv sequence without the target modified peptide. Although we could have used AlphaFold3 (62) to more properly model posttranslational modifications, we chose AlphaFold2 because at the time it was more open and could be used without restriction.

Of the four designed anti-H3K27ac mintbodies we screened, one localized in the nucleus of U2OS cells, where histone acetylation is expected to be (fig. S3A). Reassuringly, when this sequence was later input into AlphaFold3 together with its target H3 tail peptide (Fig. 3B), the model confidence (pLDDT) of the H3 peptide peaked at K27 when that residue was acetylated, even when competing acetylation marks were nearby (Fig. 3C). To more directly test specificity in an experimental setting, we tracked the cellular localization of the

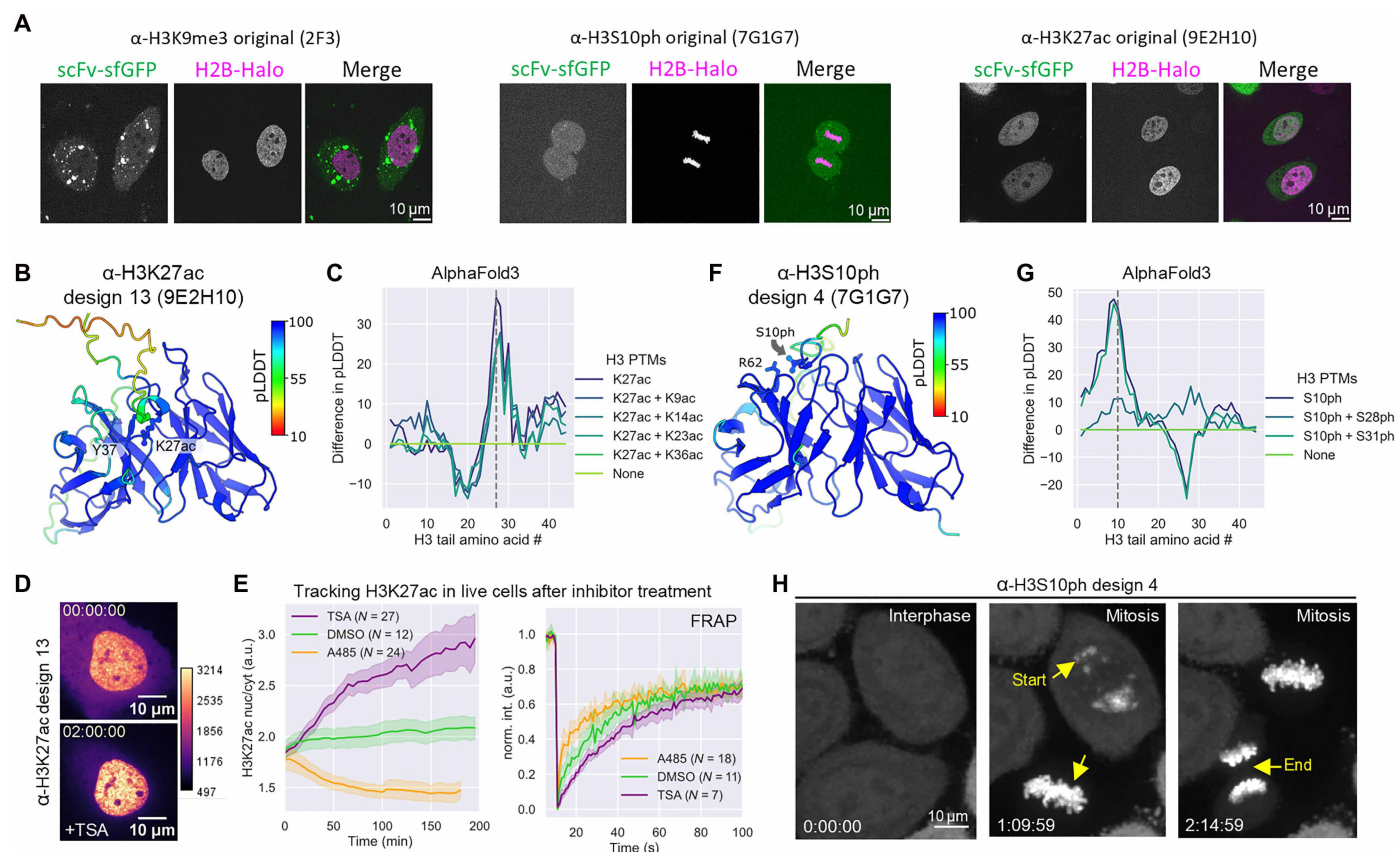


Fig. 3. Intrabodies targeting posttranslational H3 modifications. (A) Three original scFvs (green) against H3K9me3 (2F3), H3S10ph (7G1G7), and H3K27ac (9E2H10), respectively, do not colocalize with their target epitopes in the nucleus (H2B-Halo; magenta). (B) Predicted AlphaFold3 structure of design 13 of the anti-H3K27ac scFv (9E2H10) bound to the H3 tail with K27ac (color coded by pLDDT). (C) The difference in predicted pLDDT scores between AlphaFold3 models with an H3 peptide harboring acetylation [as in (B)] versus those without. The first 44 amino acids of the H3 tail are shown along the x axis, with a dashed line marking K27ac. PTM, posttranslational modification. (D) Image of a cell expressing design 13 of the anti-H3K27ac scFv pre- (top) and post-trichostatin A (TSA) treatment (bottom). (E) Left: The N/C ratio of anti-H3K27ac scFv (design 13) in living cells posttreatment with the deacetylation inhibitor TSA (purple), the vehicle control dimethyl sulfoxide (DMSO; green), or the p300 inhibitor A485 (orange). Nuclear FRAP recovery curves from cells, as in (E) (3 hours post-inhibitor treatment). (F) Predicted AlphaFold3 structure of design 4 of the anti-H3S10ph scFv (7G1G7) bound to the H3 tail with S10ph (color coded by pLDDT). (G) The difference in predicted pLDDT scores between AlphaFold3 models with an H3 peptide harboring phosphorylation [as in (F)] versus those without. The first 44 amino acids of the H3 tail are shown along the x axis, with a vertical dashed line marking S10ph. (H) Snapshots from a movie of cells expressing the anti-H3S10ph scFv (design 4) during mitosis. Scale bars, 10 μ m.

mintbody in response to the addition of the histone deacetylase inhibitor trichostatin A (TSA; to increase H3K27ac) and the p300 histone acetyltransferase inhibitor A485 (to decrease H3K27ac). As expected of a functional anti-H3K27ac intrabody, the N/C ratio increased in the presence of TSA (Fig. 3, D and E, purple, and movie S1) and decreased in the presence of A485 (Fig. 3E, orange, and movie S2). As well, FRAP recoveries were faster in TSA than they were in control cells and faster yet in A485-treated cells, indicating progressively fewer target binding sites (Fig. 3E, right, and movie S3). The FRAP recovery ($t_{1/2} \sim 15$ s) was also on par with similar measurements using purified fragmented antibodies (Fab) physically loaded into cells ($t_{1/2} \sim 21$ s) (63), indicating that the CDRs are largely intact in this mintbody.

Regarding anti-H3S10ph, four of five scFv designs localized with mitotic chromatin, as would be expected (fig. S3B). One design (design 4) had the best signal-to-noise ratio. When we input this sequence into AlphaFold3 together with its target H3 tail peptide (Fig. 3F), the model confidence (pLDDT) of the H3 peptide peaked at S10 when that residue was phosphorylated, even when competing

phosphorylation marks were nearby (Fig. 3G). This success suggested that the epitope may not be necessary in the original model used to generate the mintbody designs. Time-lapse imaging of design 4 in living cells undergoing mitosis revealed the rapid dynamics of H3S10ph and dephosphorylation (Fig. 3H). The anti-H3S10ph mintbody marked phosphorylation in patchy spots across the nucleus nearly 5 min before mitosis could be detected on the basis of H2B localization (movie S4), a result also seen using purified Fab physically loaded into cells (60). Loss of phosphorylation was also rapid as cells approached telophase, with the condensed mitotic chromatin closest to the metaphase plate harboring phosphorylation longest.

Unfortunately, all designs we tested for the anti-H3K9me3 mintbody failed to properly localize within the nucleus (fig. S3C), despite having high pLDDT scores according to AlphaFold2 and 3. However, some designs were worse than others. In particular, design 1 could not be detected in cells at all, and design 3 showed characteristic aggregates that are most likely indicative of misfolding. It remained unclear why this particular antibody suffered, although a clue may come from the parental antibody (2F3/

CMA318), from which it was difficult to generate Fab from due to the presence of protease susceptible sites in the heavy chain (52). It is possible that these sites may also hinder scFv folding, stability, and overall performance.

Developing a panel of histone modification-specific intrabodies (mintbodies)

As a final demonstration of the practicality and speed of our approach, we applied our scFv pipeline to a full panel of histone modification-specific mouse monoclonal antibodies that were previously developed, sequenced, and validated for binding specificity and strength (52, 59, 61, 64–68). We were especially curious whether there are specific histone modifications that are inherently difficult to target. Among the 19 antibody sequences we ran through the pipeline, 14 resulted in successful scFv, having at least one design that exhibited enrichment in nuclei or mitotic chromosomes (Fig. 4A and fig. S4). For example, the original anti-H4K16ac scFv failed to localize to the nucleus, but four of five scFv designs exhibited nuclear enrichment (Fig. 4B). Although the N/C ratios were high for designs 2 and 5 (Fig. 4C), their expression levels—both the intensity and the number of expressing cells—were lower than design 3, as revealed by Western blotting (Fig. 4D). Other acetylation-specific antibodies (H3K14ac, H4K5ac, H4K8ac, and H4K12ac) were also vastly improved, exhibiting strong nuclear enrichment that was completely lacking in their original counterparts (fig. S4A).

Among the 12 methylation-specific antibodies we tested, 8 localized to the nucleus, including 3 targeting monomethylation (H3K4me1 clone 19A5/CMA302 and H3K27me1 clones 4C4 and 1B3/CMA321), 2 targeting dimethylation (H3K4me2, clone 27A6/CMA303 and H3K27me2 clone 5A12/CMA322), and 2 targeting trimethylation (H3K4me3 clone 16H10/CMA304 and H3K36me3 clone 13C9/CMA333) (fig. S4C). This suggests that our inability to target H3K9me3 (fig. S3E) was most likely due to chance rather than being indicative of an inherent problem targeting methylation more generally.

Of the two phospho-specific antibodies we ran through our pipeline, one (anti-H3S10ph clone 3-5/CMA313) marked mitotic chromosomes (fig. S4B). To further test our pipeline, we also examined an scFv that was functional from the start (anti-H3T3ph clone 16B3). In this case, all five pipeline designs localized to mitotic chromosomes similar to the original (fig. S5A). Although these designs did not have superior signal-to-noise ratio compared to the original (fig. S5B), the fact that all five retained functionality highlights the robustness of our pipeline.

We hypothesized that the success of our designs stemmed from improved solubility and stability. To test this directly in a controlled setting, we expressed anti-H4K16ac mintbodies (1B2/CMA416) using an in vitro translation system. After centrifugation, the original mintbody largely precipitated, whereas the redesigned version was more evenly distributed between the supernatant and precipitate (fig. S6A). Furthermore, nearly all of the original mintbody in the

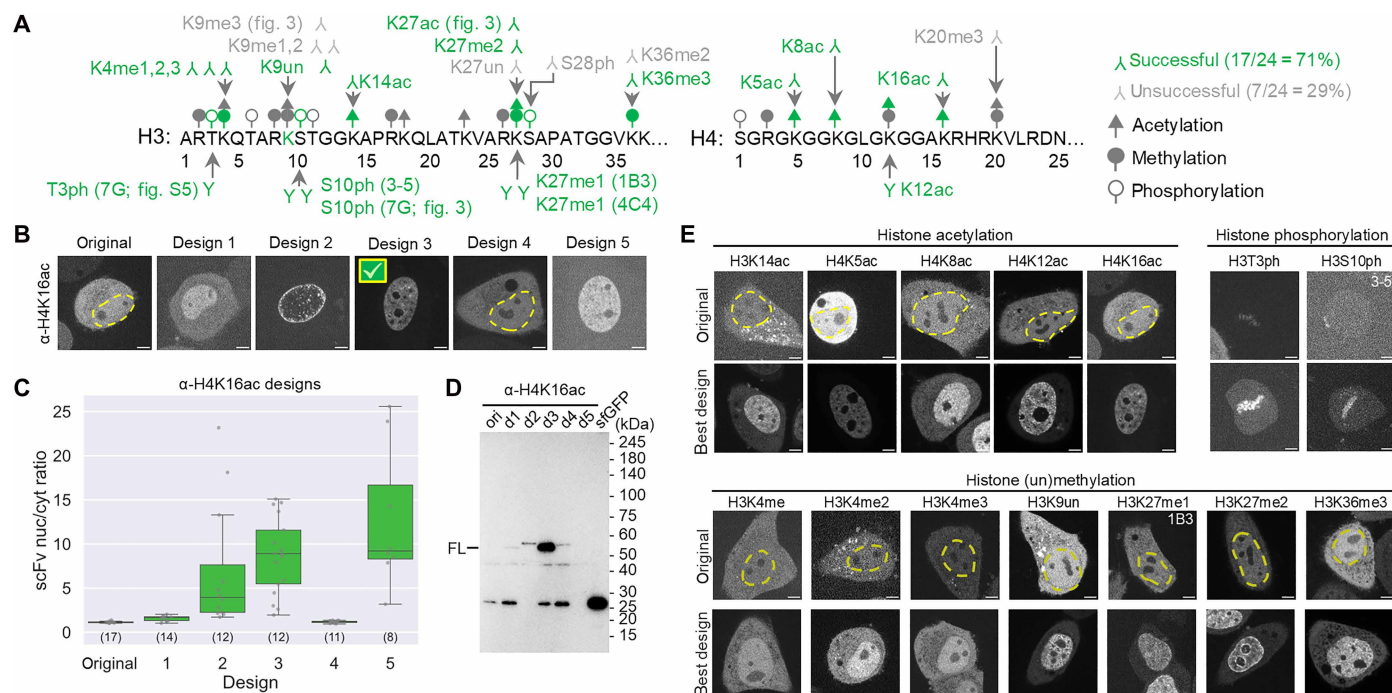


Fig. 4. A panel of intrabodies targeting diverse histone modifications. (A) Twenty-four antibodies against H3 and H4 posttranslational modifications were passed through the pipeline to generate candidate mintbodies (modification-specific intrabodies). Top-ranked (by pLDDT) candidates (four to five per antibody) were tested in living HeLa cells. Seventeen of 24 attempts were successful (highlighted in green versus gray). (B) Top: Sample images of cells expressing candidate mintbodies against H4K16ac to compare which localize better to the nucleus (which is marked by a dashed yellow line when needed). Design 3 showed the best localization pattern. (C) Sample live-cell measurement of the N/C ratio of the anti-H4K16ac mintbody designs [number of cells (N) indicated; box is 25 to 75% interquartile range; whiskers mark points within 1.5× the interquartile range]. (D) A Western blot showing that design 3 (d3) of the anti-H4K16ac mintbody has the highest full-length (FL) expression level in cells. (E) Sample live-cell images of each successful mintbody comparing the localization of the original mintbody (top rows) compared to the best design (bottom rows). The best designs localize better to the nucleus (which is marked by a dashed yellow line when needed) or, in the case of phosphorylation, to mitotic chromatin. Scale bars, 5 μ m.

supernatant precipitated after heating to 50°, 60° and 70°C, whereas a substantial fraction of the redesigned version remained soluble under the same conditions (fig. S6A). These results suggest that scFvs redesigned using our pipeline have greater thermal stability than their original counterparts.

Next, we compared the solubility of the synthesized proteins under nonreducing and reducing conditions to assess the contribution of the intramolecular disulfide bridge within the scFv. In all three original versions tested—anti-H4K16ac (1B2/CMA416), anti-H3K27ac (9E2H10/CMA309), and anti-H3K4me2 (27A6/CMA303)—the proteins tended to precipitate after centrifugation under reducing conditions, whereas they were more efficiently recovered in the supernatant under nonreducing conditions in the presence of glutathione disulfide and disulfide isomerase that facilitates disulfide bridging (fig. S6B) (69). By contrast, the designed versions were recovered predominantly in the supernatant even under reducing conditions (fig. S6B). These results suggest that the folding of the original versions is assisted by the presence of a disulfide bridge, whereas the designed versions can fold properly without it, allowing correct folding under the reducing conditions found in the cytoplasm.

To determine whether mintbodies exist as monomers or multimers, we performed fluorescence correlation spectroscopy (FCS) (70) and glycerol density gradient centrifugation using the following as size standards: superfolder GFP (sfGFP) tagged with a nuclear localization signal (NLS) [sfGFP-NLS (71); 28 kDa], Fab (~50 kDa), bovine serum albumin (BSA; 66.5 kDa), and immunoglobulin G (IgG; ~150 kDa). FCS, which measures the diffusion time of fluorescent molecules—a parameter correlated with molecular size—showed that the mintbodies, synthesized either in reducing and nonreducing conditions, diffused at a rate similar to that of Fab, falling between sfGFP-NLS and BSA, consistent with a monomeric size (fig. S6C). Similarly, in density gradient centrifugation, the mintbodies sedimented between sfGFP-NLS and BSA (fig. S6D), further supporting that they exist as monomers. These results suggest that newly designed mintbodies can form monomers in the reducing environment such as the cytoplasm.

Validating the specificity of new mintbodies

To assess the specificity of new mintbodies, we performed both *in vitro* and cellular assays. We first examined the binding specificity of three mintbodies—anti-H4K16ac (1B2/CMA416), anti-H3K27ac (9E2H10/CMA309), and anti-H3K4me2 (27A6/CMA303)—using the Modified Histone Peptide Array (Active Motif). All three tested mintbodies selectively bound to their target modifications (fig. S7A), consistent with the binding specificity of their original IgG counterparts as assessed by enzyme-linked immunosorbent assay (ELISA) (59, 67). In addition, we compared the affinity of the anti-H3K27ac (9E2H10/CMA309) mintbody with that of the monovalent Fab version of the original antibody by ELISA. Although the binding profiles obtained with two different peroxidase-conjugated secondary antibodies—anti-GFP and anti-mouse IgG (H + L)—are not directly comparable, we used high concentrations of the secondary antibodies to approach saturation levels. The designed mintbody showed slightly lower reactivity toward the H3K27ac peptide than the original Fab, but the difference was within the same order of magnitude (fig. S7B). Together with the above cell-based results (Fig. 3, D and E, and fig. S3, D to H), these findings indicate that the anti-H3K27ac mintbody can be used as a probe to measure H3K27ac levels in a manner comparable to the Fab (52).

We further characterized the specificity of selected mintbodies using live-cell assays. HeLa cells stably expressing designed mintbodies—anti-H3K14ac (7G8), anti-H4K8ac (72A9/CMA408), and anti-H4K12ac (50B3/CMA412)—were treated with TSA for 5 hours. In all cases, the N/C ratio increased upon TSA treatment (fig. S8, A and B), similar to the response observed with the designed anti-H3K27ac mintbody in Fig. 3 (D and E). These results confirm that the acetyl-specific mintbodies respond to elevated nuclear acetylation levels.

Anti-H4K5ac mintbody (4A7/CMA405) has a unique property of recognizing H4K5ac when H4K8 is unmodified, thereby preferentially detecting newly assembled H4 incorporated into chromatin during DNA replication in S phase (67). Time-lapse imaging revealed that the anti-H4K5ac mintbody colocalized with replication foci marked by proliferating cell nuclear antigen (PCNA) tagged with HaloTag, whereas the anti-H4K16ac mintbody remained in euchromatin throughout S phase (fig. S8C). Thus, the anti-H4K5ac mintbody likely retains the binding specificity of the original antibody.

We next used small-molecule inhibitors to assess the specificity of unmodified-, methyl-, and phospho-specific mintbodies. For unmodified- and methyl-specific mintbodies, HeLa cells were treated with either UNC0642, an inhibitor of G9a/GLP (72), or DS3201, an inhibitor of EZH1/2 (73), for 9 days to reduce H3K9 and H3K27 methylation levels, respectively, followed by expression of the mintbodies. The N/C ratio of the anti-H3K9un (1C6) mintbody increased in UNC0642-treated cells, consistent with the expected rise in unmodified H3K9 due to decreased H3K9me2 levels (fig. S9). The N/C ratios of anti-H3K27me1 and anti-H3K27me2 mintbodies decreased in DS3201-treated cells, where all forms of H3K27 methylation were reduced, whereas the anti-H3K36me3 mintbody was unaffected by either inhibitor.

For phospho-specific mintbodies, HeLa cells stably expressing the probes were treated with ZM447439, an inhibitor of Aurora B, or CHR6494, an inhibitor of Haspin, for 1.5 to 4.5 hours to block phosphorylation of H3S10/S28 and H3T3, respectively (fig. S10). Anti-H3S10ph mintbodies (7G1G7/CMA311 and 3-5/CMA313) localized to mitotic chromosomes in control dimethyl sulfoxide (DMSO)-treated cells and CHR6494-treated cells but not in ZM447439-treated cells (fig. S10B, top and middle). By contrast, the anti-H3T3ph mintbody (16B3) remained on mitotic chromosomes in ZM447439-treated cells—albeit at reduced levels due to partial dependence of Haspin activity on Aurora B (74)—but was absent from CHR6494-treated cells (fig. S10B, bottom). These results confirm the specificity of 7G1G7/CMA311 and 3-5/CMA313 for Aurora B-mediated H3S10ph and of 16B3 for Haspin-mediated H3T3ph. We further validated the specificity of the two anti-H3S10ph mintbodies, 7G1G7 and 3-5, whose original antibodies preferentially recognize H3S10ph adjacent to low-methylated (H3K9me0–2) and high-methylated (H3K9me2–3) states, respectively (60). These were expressed in wild-type and five H3K9 methyltransferase-knockout (5KO) immortalized mouse embryonic fibroblasts (iMEFs), which lack Suv39h1, Suv39h2, G9a, GLP, and Setdb1 and therefore lack H3K9me2 and H3K9me3 (75). While 7G1G7 localized to mitotic chromosomes in both wild-type and 5KO iMEFs, 3-5 was excluded from mitotic chromosomes in 5KO iMEFs, consistent with its requirement for H3S10ph adjacent to H3K9me2/3 (fig. S10C).

Collectively, these live-cell assays demonstrate that the designed mintbodies retain the target specificity of their original antibodies. As we have successfully established cell lines that stably express mintbodies in human (HeLa or U2OS) and mouse (NIH3T3 or iMEF)

cells, the new mintbodies expressed at an appropriate concentration do not overtly disturb cell division and viability. Mintbodies bind their targets reversibly and in a monovalent manner, which limits the duration of target engagement compared to full-length antibodies (27). However, at higher concentrations, they may still affect endogenous interactions, depending on the experimental conditions.

Analysis of mutations in successful versus unsuccessful intrabody designs

To investigate sequence features that distinguish successful from unsuccessful intrabody designs, we tabulated all sequences and ranked them by quantification in living cells (table S1). In addition, we tabulated all mutations introduced by our pipeline (table S2). This revealed a clear mutation bias inherent to the ProteinMPNN algorithm. For example, the algorithm tends to substitute residues such as glutamine (Q), arginine (R), serine (S), threonine (T), and valine (V) with lysine (K) in both successful (Fig. 5A) and unsuccessful designs (Fig. 5B). Combining all data (Fig. 5C) revealed two mutations that were common across multiple design variants: $S \rightarrow K$ and $M \rightarrow I$. Other notable mutations included $E \rightarrow G$ and $F \rightarrow E$. All other beneficial mutations (red in Fig. 5C) were observed in only a small subset of designs (<5), limiting their general predictive value.

To assess where these common mutations occurred, we mapped their positions along the globally aligned scFv coding sequence (Fig. 5, D and E, and fig. S11A). For the $S \rightarrow K$ mutation, position 82 (aligned numbering; starred in Fig. 5D) was the most predictive of success. This position corresponds to the critical residue we highlighted in fig. S2 (K80 using prealigned numbering). These data corroborate the importance of this residue, although, as we mentioned earlier, lysine at this position is not required for success. Targeted mutagenesis supports this conclusion: When we reversed the charge

at K80 (K80E) in a successful design (design 10), it did not impair its binding nor was binding significantly affected when we also mutated a neighboring residue (S79A + K80N) (fig. S12). These results suggest that specific combinations of mutations, rather than individual substitutions, are critical for intrabody functionality.

For each design, we next compared AlphaFold2-derived pLDDT, predicted template modeling (pTM), and predicted aligned error (PAE) scores, along with three other independent metrics that we computed: the solubility weighted index (SWI) (76), the log likelihood of design success from a structure-informed large language model [Evolutionary Scale Modeling–Inverse Folding 1 (ESM-IF1)] (77), and the predicted pK_a , where K_a is the acid dissociation constant, and net charge at neutral pH (table S1 and fig. S11, B and C) (78). Across most targets, successful designs had higher pLDDT, pTM, and PAE values than unsuccessful designs (16/17 cases), while SWI and log likelihood also tended to be higher for successes (14/17 cases). Predicted pK_a was less consistent, with higher values for successes in only 11/17 cases.

Individually, each metric showed some predictive capacity within specific design sets (fig. S11C). However, none provided a universal threshold applicable across all targets (fig. S11B). Metric values varied substantially between targets, indicating that design success likely depends on context-specific interactions between sequence, structure, and target epitope. These findings highlight both the promise and the current limitations of using structural and biophysical predictors for intrabody design, underscoring the need for larger datasets and integrated multimetric approaches.

DISCUSSION

In this study, we used AI-based tools to accelerate the development of functional intrabodies by modeling and optimizing their

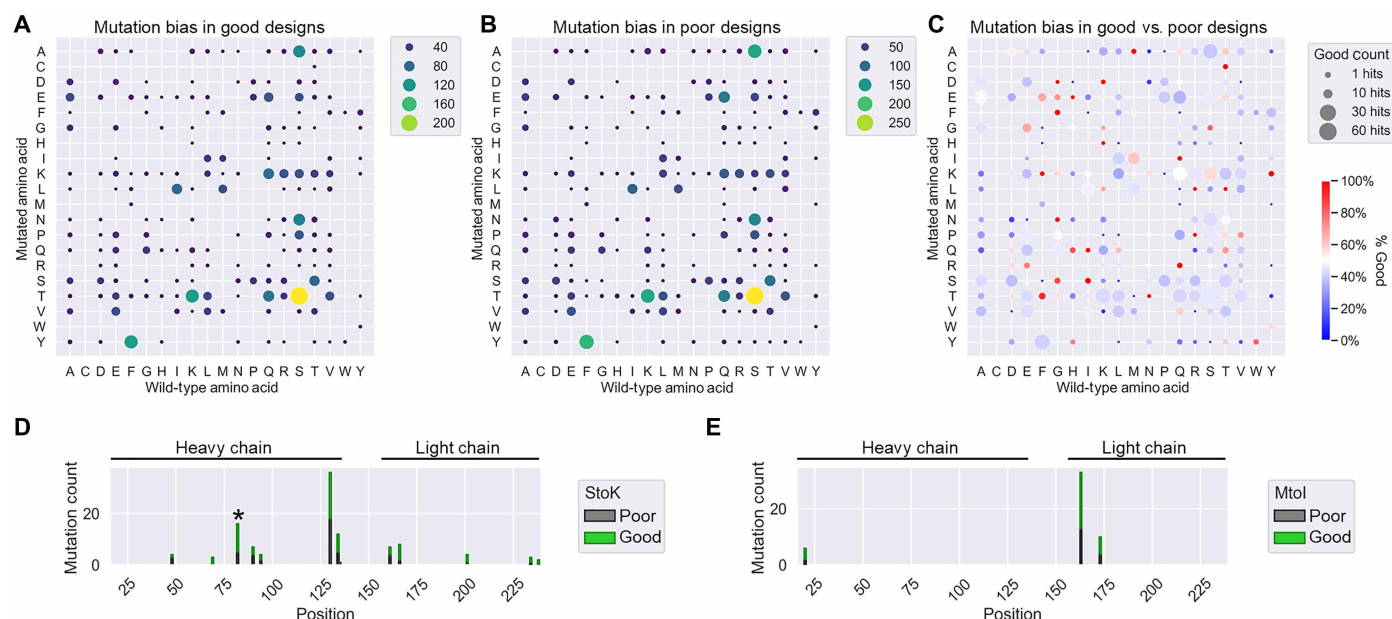


Fig. 5. Analysis of mutations in successful (good) and unsuccessful (poor) designs. (A) Table of mutations in all successful (good) intrabody designs (53 designs with $N = 3193$ mutations). The larger the dot, the more designs. (B) Mutations in all unsuccessful (poor) intrabody designs (69 designs with $N = 4331$ mutations). (C) Mutation bias in good versus poor designs. Dot size corresponds to that in (A), while color (blue to red) indicates the fraction of good designs. (D) Histogram of $S \rightarrow K$ (StoK) mutations in all designs along the scFv sequence (after alignment). The green portion of each bar indicates the number of good designs, while the gray portion indicates the number of poor designs. The star indicates position 82 (which corresponds to K80 in fig. S2). (E) Same as (D), but for $M \rightarrow I$ (Mtol) mutations.

framework regions. Much effort has been made to evolve and design CDRs to obtain antigen binders with high specificity and affinity. In contrast, although the framework regions play a critical role in securing antibody structure and stability, efforts to develop a general strategy to improve their intracellular function have been limited (14, 79). By focusing on refining these framework regions while leaving the CDRs intact, we achieved a high success rate in converting existing antibody sequences into functional intrabodies.

Of the 26 parental antibody sequences we tested, 19 were successfully converted into functional scFv intrabodies, yielding an overall success rate of ~73%. This success rate is substantially higher than the 5 to 10% success rate we have experienced when directly converting endogenous antibody sequences. We began with designs that all initially failed as scFv intrabodies (aside from anti-H3T3ph), many of which also failed after standard loop grafting onto stable scaffolds having high sequence similarity, usually an indicator of grafting success (7). Despite these difficult starting points, of a total of 127 designs the pipeline generated, 51 functioned effectively in living cells, so the average probability that any single design worked was roughly 41%.

These statistics highlight the effectiveness of our pipeline to optimize scFv framework regions for intracellular expression. As a first application, we created intrabodies against the popular synthetic FLAG tag and an epitope within the SARS-CoV-2 NP. We then focused on histone modifications, tripling the number of mintbodies available for tracking the dynamics of these elusive marks in vivo, a traditionally challenging task (11, 27). We further confirmed the quality of representative designs through detailed biochemical and live-cell validation, showing that selected mintbodies fold correctly under reducing conditions, are monomeric, and retain the target specificity of their parental antibodies both in vitro and in vivo. However, not all newly developed mintbodies have been fully characterized; therefore, their biochemical properties, such as specificity and affinity, should be evaluated for each individual application. With these intrabody sequences now stored on plasmids, it is easier to distribute them through platforms such as Addgene and to modify them, for example, to create bispecific antibodies that could be used to explore the combinatorics of the histone code hypothesis (24, 80, 81). These applications will complement existing tools such as GFP tagging, further expanding our ability to probe complex biological processes.

The potential impact of our approach is substantial. With more than 2000 solved antibody structures available in the Structural Antibody Database (82) and more than 147,000 antibody sequences from the Observed Antibody Space (83), each represents a candidate for conversion into a functional intrabody. Our pipeline requires only the antibody sequence, not the structure, and can even work without inputting the target epitope sequence. This is ideal because the number of publicly available antibody sequences is rapidly growing, as researchers prioritize preserving these sequences to address the antibody crisis caused by variability in commercial antibody batches (84, 85). Our method offers a timely approach to take advantage of this expanding resource.

Our results suggest that while AI-based tools such as AlphaFold offer valuable structural predictions, they alone are insufficient for ensuring successful intrabody design. For example, high pLDDT scores, which indicate predicted structural confidence, did not consistently correlate with functional success in our live-cell screens. This finding highlights the importance of experimental validation

and suggests a need for new metrics that better predict intrabody functionality. Notably, the failures in our pipeline were just as informative as the successes. By systematically comparing mutations across >100 successful and unsuccessful designs, we identified recurrent substitutions and positional patterns that may contribute to solubility and stability while also confirming that no single mutation guarantees success. These results emphasize that specific combinations of changes are likely required and that larger datasets will be key to revealing generalizable mutation patterns predictive of intrabody performance. In contrast, we observed that some antibodies, such as anti-H3K9me3 or anti-H3S28ph, consistently failed for reasons that remain unclear, whether due to issues with the parental antibody or other hidden factors. Despite these occasional failures, we believe that our approach will be instrumental in mapping the key residues that stabilize scFv structures, thereby improving future intrabody designs.

While our work was being prepared, new methods such as AbMPNN (86), IgMPNN (37), and AntiFold (35) have emerged, building on ProteinMPNN (42) and AlphaFold2 (29) to improve antibody structure prediction and design by incorporating antibody-specific data. Although these tools focus on optimizing CDRs to improve binding affinity (37, 86) and also prevent self-binding and nonspecific targeting (38), the extent to which these tools can also improve the framework regions of intrabodies for functionality inside living cells remains unclear. Developing tools to optimize these regions will be useful, and comparing their predictions with our experimentally validated designs will offer insights into what drives intrabody success in living cells.

To facilitate these sorts of comparisons, we provide a comprehensive list of all sequences we designed and tested, ranked by visualization and quantification in living cells (table S1). The combined sequence, mutation, and functional data now available from our study provide a valuable training set for refining design algorithms, and our validation results underscore the importance of coupling computational predictions with rigorous biochemical and cell-based testing. As more designs are generated and analyzed, this resource will grow, making it easier to distinguish successful from unsuccessful intrabody designs. Ultimately, we hope to train improved algorithms with these data, with the goal of developing a next-generation pipeline specifically trained on functional intrabodies to further reduce the need for labor-intensive live-cell screening.

MATERIALS AND METHODS

Antibody to intrabody pipeline

Our pipeline to convert antibody sequences into designed-intrabody sequences consists of five steps. First, we aligned and annotated the antibody heavy and light chain sequences, typically using Clustal (87) for alignment and ANARCI (40) or Benchling's built-in antibody property prediction tool for CDR and framework region annotation. We sometimes compared predicted CDRs and chose a conservative set of residues encompassing all predictions. Second, on the basis of the annotations from step one, we designed the scFv sequence, starting with the variable region of the heavy chain at the N terminus (containing four framework regions and three CDRs), followed by a GS-rich linker and ending in the variable region of the light chain (also containing four framework regions and three CDRs). Third, a structure was predicted for this sequence with AlphaFold2 (29) using ColabFold (41). Fourth, this structure was

used as input into ProteinMPNN (42). During the ProteinMPNN sequence design, the fixed positions included the CDRs and all residues in the scFv's framework regions that were within 3 Å of the CDRs, along with the GS-rich linker. We experimented with fewer angstroms when designing M2 anti-FLAG scFvs, but this seemed to lower the success rate (e.g., we tested 0 Å with the M2 anti-FLAG antibody, and this produced 0 successful intrabodies). By contrast, a value of 5 Å left important residues we had previously identified in the framework regions unchanged [e.g., M83, which we previously found (47) was an indicator of nonfunctional scFv]. The remaining residues form the framework of the scFv, to be redesigned by ProteinMPNN. The settings were as follows: number of sequences = 15, sampling temperature = 0.1, model = vanilla-v_48_020 or soluble-v_48_020, and backbone noise = 0.02. Fifth, we predicted the structure of all ProteinMPNN-guided sequence designs using ColabFold and ranked these by pLDDT.

We provide Python code on GitHub that implements our complete pipeline in Python (https://github.com/jbderoo/scFv_Pmpnn_AF2), using ANARCI (40) to annotate CDRs (using the Martin numbering scheme by default), LocalColabFold (41) to predict intrabody structures, and ProteinMPNN (42) to generate new framework sequences for each input. Table S3 also provides a summary of the software packages used in the design pipeline.

Plasmid construction

The nucleotide sequence of histone modification-specific antibodies was determined as previously described (88). AI-assisted scFv library designs were reverse translated into human codon optimized DNA sequences. The library scFv DNA sequences were flanked with sequences homologous to the PB533_15F11-sfGFP vector (Addgene, plasmid #167526) to be inserted at the NheI/AgeI cut sites with isothermal assembly. scFv sequences were synthesized by Twist Bioscience. In some cases, scFv sequences were synthesized by Integrated DNA Technologies (eBlocks) and cloned into the PB533 vector (System Biosciences) harboring an sfGFP sequence (57) using the In-Fusion Cloning system (Takara Bio). Plasmids for all successful scFvs will be made available at Addgene upon acceptance of the manuscript.

Stable cell line generation

Stable cell lines were generated using the PiggyBac transposon system. All scFv vectors contain PiggyBac long terminal repeats with neomycin resistance for selection. Cells were cotransfected with Super PiggyBac transposase (NovoPro or Systems Biosciences) at 1:5 transposase to transposon in a 35-mm dish at 50% confluency. Once cells reached confluency, cells were expanded into 10-cm dishes and selected for a week with G418 (750 µg/ml; or 1 mg/ml for HeLa and iMEFs). Individual colonies were handpicked for monoclonal cell lines or pooled together for polyclonal lines.

Cell culture

Human U2OS osteosarcoma or HeLa cells were grown in Dulbecco's modified Eagle's medium (DMEM; Thermo Fisher Scientific or Nacalai Tesque) supplemented with 10% (v/v) fetal bovine serum (FBS; Atlas Biologicals or Thermo Fisher Scientific), 1% (v/v) penicillin/streptomycin (Thermo Fisher Scientific or Sigma-Aldrich), and 1 mM L-glutamine (Thermo Fisher Scientific or Sigma-Aldrich) at 37°C in 5% CO₂. Mouse fibroblast NIH3T3 cells were grown in DMEM (American Type Culture Collection, 30–2002) supplemented with 10% (v/v) FBS (EqualFETAL, Atlas Biologicals) and 1% (v/v)

Pen/Strep (Gibco, 15140122) at 37°C in 5% CO₂. Wild-type and Suv39h1/Suv39h2/Setdb1/G9a/GLP 5KO iMEFs (75) were grown in DMEM (Nacalai Tesque) supplemented with 10% (v/v) FBS (Thermo Fisher Scientific), 1% (v/v) Pen/Strep (Sigma-Aldrich), 1 mM L-glutamine (Sigma-Aldrich), 1% nonessential amino acids (Fujifilm; 100×), 1% 2-mercaptoethanol (Fujifilm; 10 mM stock), and 1% sodium pyruvate (Thermo Fisher Scientific; 100 mM stock) at 37°C in 5% CO₂. Trypsinization for cell passaging was performed using TrypLE Express (Thermo Fisher Scientific, 12605010) or 0.25% trypsin/0.02% EDTA in phosphate-buffered saline (PBS; Nacalai Tesque).

Transfection

The day before imaging, cells were seeded on 35-mm MatTek chambers or 24-well glass-bottom plates (AGC Techno Glass) at ~70% confluency and transfected. Transfection was performed using Lipofectamine LTX or 2000 (Thermo Fisher Scientific) with 0.8- to 1-µg total plasmid DNA. Cells were washed three times with DMEM the next morning. In experiments using HaloTag, staining was performed using 200 nM of Janelia Fluor 646 HaloTag Ligand for 30 min and then washed three times with DMEM. Cells were then kept in supplemented DMEM lacking phenol red or FluoroBrite (Thermo Fisher Scientific) containing 10% FBS for imaging.

For colocalization imaging experiments (Figs. 1, B and E, and 2, A and D; and figs. S1, A and D, and S2A), the cells were seeded on 35-mm MatTek chambers with 70% confluency. The next day, 1.25 µg of the scFv-GFP construct and 1.25 µg of the epitope-tagged mCh-H2B construct were transfected into cells using Lipofectamine LTX. For fig. S1D, Mito-mCh-1xFLAG (46) was used instead of epitope-tagged mCh-H2B. The medium was changed to complete DMEM medium at 3 hours posttransfection. The cells were imaged at 20 hours posttransfection. Cell treatment for anti-H3K27ac design 13 imaging experiments was performed as above except 2.0 µg of scFv-GFP and 1.0 µg of mCh-H2B constructs were used (Fig. 3, D and E).

In vitro translation and downstream analysis

Templates for in vitro transcription-translation coupled systems were generated by polymerase chain reaction, and their sequence information is summarized in table S4. For fig. S6A, mintbodies were produced by an in vitro transcription-translation coupled system using PUREfrex 2.0 (Gene Frontier). Approximately 90 ng of each template DNA in a 20-µl reaction mixture was incubated for 6 hours at 37°C. After centrifugation at 12,000g for 30 min at 4°C, supernatant was transferred to a new tube. After incubating supernatant fractions for 30 min at 50°, 60°, and 70°C, precipitated materials were removed by centrifugation as above. All fractions were added with an equal volume of 2× SDS sample buffer and heated for 10 min at 95°C for Western blotting.

For figs. S6 (B to D) and S7, mintbodies were produced using the Cell-Free Protein Expression kits Musabou Kun "N100," "N100 SS," or "N1000" (Taiyo Nippon Sanso and FujiFilm) according to the manufacturer's instructions. N100, which contains dithiothreitol, was used under reducing conditions; N100 SS, which contains glutathione disulfide and the disulfide bond isomerase C, was used under nonreducing conditions to promote disulfide bridge formation in the synthesized protein; and N1000 was a large-scale version of N100. For the N100 and N100 SS reactions, ~250 ng of each template DNA was added to a 100-µl reaction mixture and incubated for 16 to 24 hours at 30°C with gentle shaking. After collecting

reaction mixture, the dialysis unit was washed with 100 μ l of PBS, and the remaining solution was collected. After centrifugation of 140 μ l of mixture at 12,000g for 30 min at 4°C, 130 μ l of supernatant was transferred to a new tube, and the precipitate was resuspended in 130 μ l of PBS. For Western blotting, 1 μ l of each fraction was mixed to achieve 1 $\times$ concentration with 2 $\times$ SDS sample buffer and PBS and heated for 10 min at 95°C.

A fraction (20 μ l) of supernatant was analyzed using the Smart FCCS System (Cosmo Bio) to measure concentration and mobility based on GFP fluorescence excited with a 488-nm laser (100% output). The following were used as size standards: sfGFP-NLS (71), Alexa Fluor 488–labeled Fabs (52, 67), Alexa Fluor 488–labeled BSA (Fraction V, Nacalai Tesque), and Alexa Fluor 488–labeled IgG (59, 67). Diffusion time was determined by fitting the correlation curves to a single-component model using the built-in Smart FCCS software.

For glycerol gradient centrifugation, 30 μ l of supernatant was loaded onto a 10 to 30% glycerol gradient [4.5 ml in 100 mM potassium acetate, 30 mM KCl, 10 mM Na₂HPO₄, 1 mM dithiothreitol (DTT), 1 mM MgCl₂, and 1 mM adenosine 5'-triphosphate (ATP)]. After centrifugation at 45,000 rpm for 20 hours at 4°C in a Beckman SW 55 Ti rotor, 18 fractions (0.25 ml each) were collected, and every other fraction was analyzed by SDS–polyacrylamide gel electrophoresis, followed by immunoblotting. Input sample was 10-fold diluted by 10% glycerol (in 100 mM potassium acetate, 30 mM KCl, 10 mM Na₂HPO₄, 1 mM DTT, 1 mM MgCl₂, and 1 mM ATP). sfGFP-NLS (71), BSA (Fraction V, Nacalai Tesque), and IgG (59) were used as size standards and analyzed by Coomassie Brilliant Blue (CBB) staining with Bullet CBB Stain Lite (Nacalai Tesque).

The specificity of in vitro–produced mintbodies was assessed using the MODified Histone Peptide Array with MODified Array Labeling Kit (Active Motif) according to the manufacturer's instructions. After blocking with Blocking Buffer AM2 (Active Motif), the array slides were incubated with mintbody (20 to 600 ng/ml) in a 3-ml Blocking Buffer AM2 for 20 hours, washed three times with 1 $\times$ Wash Buffer AM4 (Active Motif) (5 min each), incubated with 1000 $\times$ diluted peroxidase-conjugated anti-GFP (MBL) in a 3-ml Blocking Buffer AM2 for 30 min, and washed again three times with 1 $\times$ Wash Buffer AM4 (2 min each) before chemiluminescence detection using enhanced chemiluminescence reagents (Active Motif).

The affinity of the anti-H3K27ac mintbody was compared with that of the Fab prepared from the original IgG antibody (52) by ELISA. Microtiter plates were coated with streptavidin (100 μ g/ml; Fujifilm) at room temperature overnight, followed by coating with 1 μ g/ml of biotinylated H3K27ac or control H3K27 peptide (Active Motif) at room temperature overnight. After washing with PBS containing 0.05% Tween 20 (PBST), the plates were incubated for 30 min with 1% BSA in PBST and then with a one-third dilution series of mintbody or Fab in PBST containing 1% BSA, starting at 1 μ g/ml, for 20 hours. Plates were washed three times with PBST and incubated with peroxidase-conjugated secondary antibody—anti-GFP (MBL) or anti-mouse IgG (H + L) (Jackson ImmunoResearch)—in PBST containing 1% BSA at 1:50 or 1:100 dilution for 30 min. After three washes with PBST, o-phenylenediamine-2HCl (0.26 mg/ml; Fujifilm) in 0.1 M citrate buffer (pH 5.0) containing 0.01% hydrogen peroxide (Fujifilm) was added, and absorbance at 450 nm was measured. Background absorbance from the H3K27 peptide was subtracted, and the resulting values for the H3K27ac peptide were plotted.

Immunoprecipitation

For fig. S1D, HEK293T cells were seeded at 80% confluency in a 60-mm dish and transfected with 3.5 μ g each of GFP-tagged anti-FLAG scFv (design 12) and FLAG-tagged mCh using Lipofectamine 3000. Following 20 hours posttransfection, cells were lifted with 1 $\times$ PBS with 1 mM EDTA and pelleted at $\times$ 300g for 5 min. Cell pellets were resuspended in 200 μ l of ice-cold lysis buffer [10 mM tris/Cl (pH 7.5), 150 mM NaCl, 0.5 mM EDTA, 1 $\times$ protease cocktail inhibitor, and 0.01% Tween-20]. Lysis was performed for 30 min and extensively pipetted every 10 min. After removing cell debris by centrifugation at $\times$ 1700g for 10 min at 4°C, lysates were immunoprecipitated using GFP-Trap Magnetic Agarose (ChromoTek). First, 25 μ l of bead slurry was equilibrated three times with 500 μ l of ice-cold dilution buffer [10 mM tris/Cl (pH 7.5), 150 mM NaCl, and 0.5 mM EDTA]. After the supernatant was removed, the cell lysates mixed with 300 μ l of dilution buffer were added to the beads. Lysate and bead mixture were rotated overnight at 4°C. Beads were washed three times using 500 μ l of dilution buffer. To elute, beads were mixed with 80 μ l of 2 $\times$ SDS-sample buffer and boiled for 10 min at 95°C. The supernatant was then used for Western blot analysis along with cell lysate.

Western blotting

For fig. S1D, cell lysates and elution from immunoprecipitation were loaded on a polyacrylamide gel (10% Bis/tris with 4% stacking gel) and transferred onto polyvinylidene difluoride (PVDF) membrane (125 V for 1 hour). The membrane was blocked with 2% Casein 1 $\times$ TBST [20 mM tris-HCl (pH 8.0), 150 mM NaCl, and 0.05% Tween 20] for 1 hour. The first primary staining was done with mouse anti-FLAG (1:500; FUJIFILM Wako), incubated for 2 hours at room temperature, and then washed three times with 1 $\times$ TBST. The second primary staining was done with rabbit anti-GFP (1:500; Life Technologies or Thermo Fisher Scientific), incubated overnight at 4°C, and then washed three times with 1 $\times$ TBST. Secondary staining was done with a mixture of goat anti-rabbit IR700 and goat anti-mouse IR800 (1:2000 each; Rockland), incubated for 1 hour at room temperature, and then washed three times with 1 $\times$ TBST.

For Fig. 4D, cells were plated on 12-well plates (Corning, 3513) at 1.0×10^5 cells per well and transfected with 1.6 μ g of plasmid DNA using Lipofectamine 2000 (Thermo Fisher Scientific). The next day after transfection, cells were washed three times with PBS and lysed with 1 $\times$ SDS-gel loading buffer [2% SDS, 10% glycerol, 100 mM tris-HCl (pH 6.8), 100 mM dithiothreitol, and 0.01% bromophenol blue; 150 μ l per well], before incubating at 95°C for 5 min.

Protein samples were separated in polyacrylamide gels (5 to 20%, 17 well; SuperSep Ace, FUJIFILM Wako) and transferred onto a 90 mm–by–90 mm PVDF membranes (FluoroTrans W, FUJIFILM Wako) using a semi-dry system (Trans-Blot Turbo, Bio-Rad; 1.3 A for 15 min) with EzFastBlot (Atto) as a transfer buffer. Membranes were washed with TBST [20 mM tris-HCl (pH 8.0), 150 mM NaCl, and 0.05% Tween 20], blocked for 20 min in Blocking One (Nacalai Tesque) with gentle shaking, and washed with TBST. Membranes were then incubated with mouse anti-GFP antibody (GF200, Nacalai Tesque; 1:2000 dilution) in Can Get Signal Solution 1 (TOYOBO) for 1 hour at room temperature, washed three times with TBST for 5 min each, incubated with horseradish peroxidase-conjugated sheep anti-mouse Ig (Jackson ImmunoResearch; 1:10,000 dilution) in the Can Get Signal Solution 2 (TOYOBO) for 1 hour at room

temperature, and washed three times with TBST for 5 min each. Chemiluminescence signals were developed using ImmunoStar LD (FUJIFILM Wako) and detected using a luminescent imager LumiGraph II (Atto).

Immunofluorescence

For fig. S3 (D to F), U2OS cells were seeded on 35-mm MatTek chambers and transfected with 1 μ g of GFP-tagged anti-H3K27ac scFv. The following day, the medium was removed, and cells were washed. All washes were done with 1 $\times$ PBS three times. Cells were then fixed and slightly permeabilized with 4% paraformaldehyde and 1% Triton X-100 in 1 $\times$ PBS for 10 min and washed. Fixed cells were fully permeabilized with 1% Triton X-100 in 1 $\times$ PBS for 20 min and washed. The cells were then blocked using 10% Blocking One-P (Nacalai Tesque) in 1 $\times$ PBS for 20 min and washed. For primary antibody staining, cells were incubated in a 1:3000 anti-H3K27ac (9E2H10) dilution in 1 $\times$ PBS for 1 hour at room temperature and washed. For secondary antibody staining, cells were incubated in a 1:2000 anti-mouse Alexa Fluor 647 dilution in 1 $\times$ PBS for 1 hour at room temperature in the dark and washed. To stain DNA, cells were incubated in 300 μ M 4',6-diamidino-2-phenylindole (DAPI) in 1 $\times$ PBS for 5 min in the dark and washed. Cells were kept in 1 $\times$ PBS and sealed for imaging.

Cell imaging

Olympus spinning-disk confocal microscope

For Figs. 1 (B and E to G), 2 (A and D to F), and 3 (D to E); figs. S1A (top), S2A, and S3A (right); and movie S2, images were captured on an Olympus IX8 spinning-disk confocal microscope (CSU-22 head) equipped with an iXon Ultra 888 EMCCD camera (Andor) operated by SlideBook (Intelligent Imaging Innovations) coupled with a Phasor photomanipulation unit (Intelligent Imaging Innovations) and a 60 $\times$ [numerical aperture (NA): 1.42] oil-immersion objective. Cells were enclosed in a custom-built 37°C chamber with 5% CO₂ throughout imaging. A 488-nm laser was used to capture GFP-tagged scFvs, and a 561-nm laser was used to capture H2B-mCh.

Imaging for Figs. 1F and 2E were acquired at full cell volumes with 17 z-stacks at 0.5- μ m step size with the 488-nm laser set to 50% power and 100 ms and the 561-nm laser at 30% power with 100-ms exposure. Time-lapse imaging in Fig. 3E (left) was captured with 21 z-stacks at 0.5- μ m step size and 5-min interval for 175 to 200 min. The 488-nm laser was set to 40% power at 200-ms exposure time, and the 561-nm laser was set to 20% power at 100-ms exposure time. FRAP experiments (Figs. 1G, 2F, and 3E, right) were performed on cells transiently transfected (as described above). Single-plane imaging was performed about the center of the nucleus. Before photobleaching, 10 frames were acquired at 1-s time interval. Between frames 10 and 11, the Phasor 488-nm laser was set to bleach a circle of $\sim$ 1 μ m in diameter for 100 ms at 90% power. Image captures continued every 1 s for an additional 170 to 250 time points.

For drug treatment imaging and FRAP of U2OS cells expressing anti-H3K27ac mintbody (Fig. 3, D and E), the position of cells expressing both the scFv and mCh-H2b was recorded, and an initial image was taken. Cells were imaged in 2.5 ml of medium.

A total of 0.5 ml of medium was removed and mixed with 2 μ l of drug in DMSO or DMSO only and then added back to the imaging dish. Working:final concentrations of TSA and A485 were 625 μ M:500 nM and 12.5 mM:10 μ M, respectively. Starting 5 min after drug addition, the cells were imaged every 5 min for 2 to 3 hours.

Cells, which were not imaged during the 3-hour time lapses, were used for the FRAP experiments as described in the previous section.

For fixed cell imaging in fig. S3 (D to F), full nuclear volumes were taken with 20 z planes at 0.5- μ m steps using the 100 $\times$ oil immersion objective with the 405-, 488-, and 640-nm lasers set to 100% power, and exposure times of 100, 2500, and 300 ms were used, respectively.

Custom HILO microscope

For figs. S1A (bottom) and S3A imaging, a custom-built high-inclined thin-illumination fluorescence microscope (89) equipped with a 60 $\times$ (NA: 1.49; Olympus) oil-immersion objective lens, three solid-state laser lines (488, 561, and 637 nm; Voltran), a T660lpxr ultraflat imaging grade emission image splitter (Chroma), and two iXon Ultra 888 EMCCD cameras (Andor) was used. A 300-mm achromatic lens (Thorlabs) was used to focus images onto the camera instead of the standard 180-mm Olympus tube lens, resulting in an effective magnification of $\times$ 100. Cells were incubated in a 37°C chamber with 5% CO₂ (Okolab). Images were captured at 512 pixel-by-512 pixel size and 53.64-ms exposure time. A single z plane was captured about the center of the nucleus.

Dragonfly and Yokogawa CSU-W1 microscopes

For Fig. 3A and figs. S3 to S5, S8, and S9, a spinning-disk confocal system (Andor Dragonfly or Yokogawa CSU-W1 for figs. S3B and S8C) with an inverted microscope (Ti-E, Nikon), operated by Fusion software version (ver.) 2.2.0.50 (Andor) or NIS-Elements software ver. 5.11.03 (Nikon; for Fig. 3H and fig. S3B), was used with a heated stage at 37°C supplemented with 5% CO₂ (Tokai Hit). Single optical sections were acquired with a 100 $\times$ Plan Apo VC (NA: 1.4) oil-immersion objective lens, a 405/488/561/640 or a 405/470/555/640NIR (for fig. S3B) dichroic mirror, a 521/38 emission filter or a 520/60 emission filter (for fig. S3B), a laser unit (5 to 20% transmission of 488 nm; LC-ILE-400-M, Andor; or 30% transmission of 470 nm with a 10% neutral density filter; LDI-7 Laser Diode Illuminator; Chroma Technology, Japan, for fig. S3B), and an EMCCD (iXon Ultra, Andor; gain multiplier: 300, exposure time: 0.3 to 0.5 s; or iXon+, Andor; gain multiplier: 300, exposure time: 0.1 to 1 s for fig. S3B). Transmission images were also acquired.

For time-lapse imaging (Fig. 3H and movie S4), HeLa cells stably expressing sfGFP-tagged anti-H3S10ph (7G1G7) scFv (design 4) and H2B-Halo (55) were plated on a 35-mm glass-bottom dish (AGC Technology Solutions). The next day, the cells were stained with 100 nM Janelia Fluor 646 HaloTag Ligand (Promega) for 1 hour, before replacing the medium with FluoroBrite (Thermo Fisher Scientific) containing FBS (Thermo Fisher Scientific) and setting onto a heated stage at 37°C supplemented with 5% CO₂ (Tokai Hit) featured on a spinning-disk confocal system (Yokogawa CSU-W1) with an inverted microscope (Ti-E, Nikon), operated by NIS-Elements software ver. 5.11.03 (Nikon). Images were acquired with a 100 $\times$ Plan Apo VC (NA: 1.4) oil-immersion objective lens, a 405/470/555/640NIR dichroic mirror, 520/60 and 690/50 emission filters, a laser unit (20% transmission of 470 nm and 15% of 640 nm with a 10% neutral density filter; LDI-7 Laser Diode Illuminator, Chroma Technology, Japan), and an EMCCD (iXon+, Andor; gain multiplier: 300, exposure time: 500 ms). Fifteen different focal planes were imaged at 1- μ m intervals every 5 min. Max intensity projection images are shown in Fig. 3H and movie S4. For fig. S8C, Halo-PCNA (55) was transiently expressed in HeLa cells that stably express anti-H4K5ac (4A7 d5; design 5). Single-section images were acquired every 20 min.

Evident SpinSR confocal microscope

For fig. S10, iMEFs were incubated with Hoechst 33342 (100 μ g/ml; Thermo Fisher Scientific) in FluoroBrite (Thermo Fisher Scientific) containing FBS (Thermo Fisher Scientific) for 30 min and then with 0.5 μ M ZM447439 (1 mM DMSO stock; Tocris Bioscience), 50 nM CHR-6494 (0.1 mM DMSO stock; MedChemExpress), or 0.05% DMSO (Nacalai Tesque) in the same medium for 1.5 to 4.5 hours. Cells were imaged using an IXplore SpinSR (Evident), equipped with a heated stage (Tokai Hit) maintained at 37°C with 5% CO₂, under confocal mode operated via cellSens Dimension 3.1 software (Evident), using a PLAPON 60 $\times$ OSC2 oil-immersion objective lens (NA: 1.4). Images were collected using 405- and 488-nm lasers (OBIS, Coherent; 20% of 100 mW each, 500 ms exposure), with a 405/488/561/640 dichroic mirror, B447/60 and B525/50 emission filters, and a scientific complementary metal-oxide semiconductor camera (ORCA-Flash4.0, Hamamatsu Photonics).

Image quantification

Nuclear/chromatin to cytoplasmic ratio

The ratio of nuclear to cytoplasmic signal intensity of cells from movies [Figs. 1F, 2E, and 3E (left) and fig. S12] was calculated using custom scripts [Nuc2Cyt scripts on (<https://github.com/Colorado-State-University-Stasevich-Lab/>)]. The maximum intensity projection of image stacks of full cell volumes was used for quantification. Background fluorescence was removed by subtracting the first percentile of intensity values, and negative values were set to zero. Nuclei were identified using Otsu thresholding of H2B, and a binary mask was created to measure the nuclear signal. To measure cytoplasmic signals, a ring mask was generated around each nucleus with a 6-pixel width. The average intensity of intrabodies for both nuclear and cytoplasmic regions was measured, and the ratios were then calculated. For live-cell tracking experiments (Fig. 3, D and E, left), each cell in a time lapse was given a unique identifier using Trackpy, which was used to link the nuclear and cytoplasmic signals of individual cells for the duration of the experiment.

For Fig. 4C and figs. S3 to S5, nuclear and cytoplasmic areas in mintbody-expressing cells were selected manually, assisted with transmission images of cells. After the background intensity in a cell-free area was subtracted from the nuclear and cytoplasmic intensities, the nuclear to cytoplasm intensity ratio was determined.

For fig. S10, rectangular areas containing chromosomes in mitotic cells were selected, and chromosome regions were detected on the basis of Hoechst staining with Huang thresholding and binary close function (to cover the entire chromosome areas without gap) using ImageJ-Fiji 1.54g (<http://imagej.org>) to measure intensities in chromosome and outside (cytoplasm) regions. After background subtraction, the ratio of chromosomes to cytoplasm was determined.

Line profile Pearson correlation analysis

For fig. S3F, line profile correlations were calculated using a custom script. Signal intensity values along a line spanning about half of the nuclear interior on a single *z* plane were used to measure Pearson coefficients between GFP-tagged anti-H3K27ac (design 13), anti-H3K27ac (parental antibody), and DAPI. For the representative profile graph in fig. S3E, the line spanned beyond the nuclear boundaries to emphasize colocalization, but the region outside of the nucleus was not used in the Pearson coefficient calculation.

Fluorescence recovery after photobleaching

FRAP curves (Figs. 1G and 2F) were calculated using a custom script written in Python (https://github.com/Colorado-State-University-Stasevich-Lab/frap_analysis). Briefly, the FRAP spot location for the measurement was determined by subtracting the first post-FRAP frame (frame 11) from the previous frame (frame 10). The resulting image was used to create a mask that marked the bleached area, and a difference-of-Gaussians filter was applied to further enhance the precise location of the center of the FRAP spot. A circular region was then inscribed into the bleached area, and the average signal from this region was plotted for the duration of the experiment (normalized to the intensity from frames 1 to 10).

ChatGPT

To improve the readability and clarify of the manuscript, ChatGPT version 4 was used in some cases with the prompt “suggest grammatical revisions to improve clarity.” All generated content was carefully reviewed, edited, and approved by the authors.

Supplementary Materials

The PDF file includes:

Figs. S1 to S12
Legends for movies S1 to S4
Legends for tables S1 to S4

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S4
Tables S1 to S4

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Acknowledgments: We thank P. S. Karupiah and J. Leavitt for assistance with preliminary experiments during the rotation in the Stasevich Lab. We thank the Jennifer DeLuca Lab (Colorado State University) for scientific discussions. We thank R. Owen from the DeLuca Lab for providing an anti-GFP antibody. We thank R. Jeep from the Chaoping Chen laboratory (Colorado State University) for providing Western blot reagents. We thank H. Ueno (Tokyo Institute of Technology) for technical assistance and the Integrative Bioscience Facility at Institute of Science Tokyo for DNA sequencing. We acknowledge the use of ChatGPT version 4 to improve the clarity of the manuscript. **Funding:** T.J.S., G.G., G.F., T.M., H.P.F., and R.H. were supported by the National Institutes of Health (NIH) R35GM119728. T.J.S., C.D.S., J.D., C.D.S., and B.J.G. were supported by NIAID R01AI168459. H.K. was supported by the Japan Society for the Promotion of Science (JSPS) KAKENHI (JP17H01417, JP21H04764, and JP24H02325), the Japan Agency for Medical Research and Development (AMED) Basis for Supporting Innovative Drug Discovery and Life Science Research (BINDS; JP24ama121020) and ASPIRE (24jf0126008h0001), and the Shimadzu Corporation. D.M. was supported by the Japan Science and Technology Agency (JST) Support for Promoting Research Initiated by the Next Generation (SPRING) (JPMJSP2180). Y.S. was supported by JST CREST (JPMJCR2056). Y.O. was supported by JSPS KAKENHI (JP23H00372, JP24H02323, and JP24K21949), AMED BINDS (JP24ama121017j0001) and ASPIRE (24jf0126008h0001), MEXT Promotion of Development of a Joint Usage/Research System Project: Pan-Omics DDRIC, MRCI for High Depth Omics, CURE:JPMXP1323015486 for MIB and RiIT in Kyushu Univ. N.Z. was supported by the NIH R00GM141453 and the Cystic Fibrosis Foundation grant 005749A123. S.R.B. and S.G. were supported by the National Science Foundation (2236710). **Author contributions:** Conceptualization: D.M., B.J.G., H.K., and T.J.S. Methodology: D.M., G.F., H.P.F., R.H., S.G., C.D.S., Y.O., Y.S., H.K., and T.J.S. Software: D.M., J.D., R.H., C.D.S., H.K., and T.J.S. Validation: G.G., D.M., S.R.B., G.F., H.P.F., R.H., and T.J.S. Formal analysis: G.G., D.M., J.D., S.R.B., R.H., H.K., and T.J.S. Investigation: G.G., D.M., S.R.B., G.F., T.M., H.P.F., R.H., N.Z., E.H.M., H.K., and T.J.S. Resources: G.G., R.H., N.Z., S.G., C.D.S., Y.O., Y.S., H.K., and T.J.S. Data curation: G.G., J.D., and T.J.S. Writing—original: G.G., J.D., R.H., B.J.G., H.K., and T.J.S. Writing—review and editing: G.G., D.M., S.R.B., G.F., H.P.F., S.G., C.D.S., B.J.G., H.K., and T.J.S. Visualization: G.G., S.R.B., C.D.S., Y.S., and T.J.S. Supervision: R.H., C.D.S., B.J.G., H.K., and T.J.S. Project administration: C.D.S., B.J.G., and T.J.S. Funding acquisition: N.Z., S.G., C.D.S., B.J.G., Y.O., Y.S., H.K., and T.J.S. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** Code for generating intrabody designs from antibody sequences is available on GitHub (https://github.com/jbderoo/scFv_Pmpnn_AF2). Code for image analysis, including FRAP and nuclear-to-cytoplasm measurements, is also available on GitHub (https://github.com/Colorado-State-University-Stasevich-Lab/frap_analysis). Table S1 provides a listing of all generated intrabody sequences, along with their nuclear-to-cytoplasmic expression ratios. All code and data are also available for download on FigShare (<https://doi.org/10.6084/m9.figshare.29972029>).

Submitted 31 March 2025
Accepted 4 December 2025
Published 2 January 2026
10.1126/sciadv.adx8352

AI-assisted protein design to rapidly convert antibody sequences to intrabodies targeting diverse peptides and histone modifications

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Sci. Adv. **12** (1), eadx8352. DOI: 10.1126/sciadv.adx8352

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